CSE5DMI

WEEK 02

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LA TROBE UNIVERSITY

OUTLINE

ATTRIBUTES AND OBJECTS

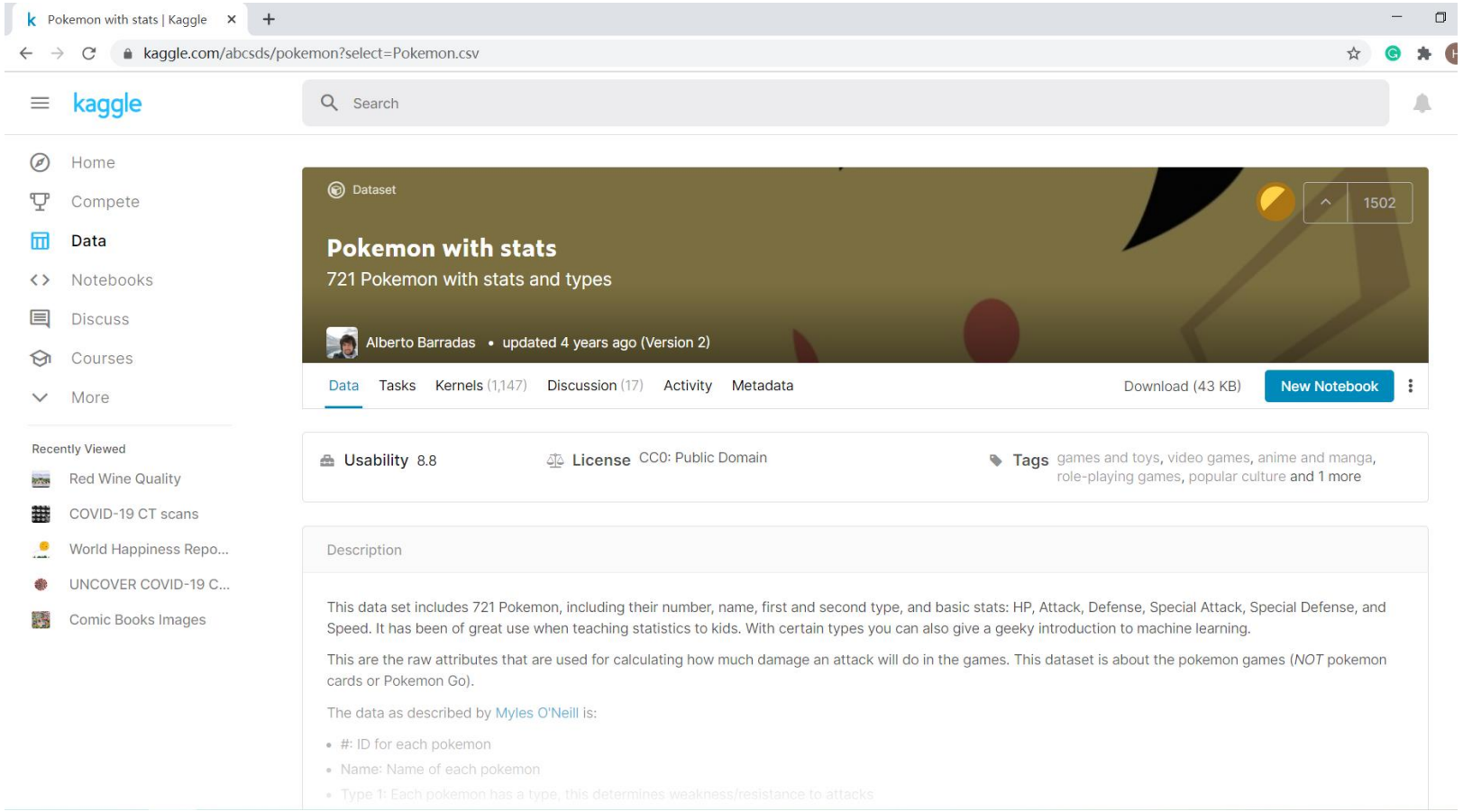
TYPES OF DATA

DATA QUALITY

SIMILARITY AND DISTANCE

DATA PREPROCESSING

WHAT IS DATA?



https://www.kaggle.com/datasets/abcsds/pokemon

What information can you identify from this dataset?

The aim is not memorising definitions only.
We repeatedly ask: **what does this data representation** allow us to do in a mining task?

Good data mining begins before the algorithm.

WHAT IS DATA?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	#	Name	Type 1	Type 2	Total	HP	Attack	Defense	Sp. Atk	Sp. Def	Speed	Generatio	Legendary	
2	1	Bulbasaur	Grass	Poison	318	45	49	49	65	65	45	1	FALSE	
3	2	Ivysaur	Grass	Poison	405	60	62	63	80	80	60	1	FALSE	
4	3	Venusaur	Grass	Poison	525	80	82	83	100	100	80	1	FALSE	
5	3	VenusaurM	Grass	Poison	625	80	100	123	122	120	80	1	FALSE	
6	4	Charmand	Fire		309	39	52	43	60	50	65	1	FALSE	
7	5	Charmelec	Fire		405	58	64	58	80	65	80	1	FALSE	
8	6	Charizard	Fire	Flying	534	78	84	78	109	85	100	1	FALSE	
9	6	CharizardM	Fire	Dragon	634	78	130	111	130	85	100	1	FALSE	
10	6	CharizardM	Fire	Flying	634	78	104	7						
11	7	Squirtle	Water		314	44	48	6						
12	8	Wartortle	Water		405	59	63	8						
13	9	Blastoise	Water		530	79	83	10						
14	9	BlastoiseM	Water		630	79	103	12						
15	10	Caterpie	Bug		195	45	30	3						
16	11	Metapod	Bug		205	50	20	5						
17	12	Butterfree	Bug	Flying	395	60	45	5						
18	13	Weedle	Bug	Poison	195	40	35	3						
19	14	Kakuna	Bug	Poison	205	45	25	5						
20	15	Beedrill	Bug	Poison	395	65	90	4						
21	15	BeedrillMe	Bug	Poison	495	65	150	4						
22	16	Pidgey	Normal	Flying	251	40	45	4						
23	17	Pidgeotto	Normal	Flying	349	63	60	5						
24	18	Pidgeot	Normal	Flying	479	83	80	7						
25	18	PidgeotMe	Normal	Flying	579	83	80	8						
26	19	Rattata	Normal		253	30	56	3						
27	20	Raticate	Normal		413	55	81	6						
28	21	Spearow	Normal	Flying	262	40	60	3						
29	22	Fearow	Normal	Flying	442	65	90	6						
30	23	Ekans	Poison		288	35	60	4						
31	24	Arbok	Poison		438	60	85	6						
32	25	Pikachu	Electric		320	35	55	4						
33	26	Raichu	Electric		485	60	90	5						
34	27	Sandshrew	Ground		300	50	75	8						
35	28	Sandslash	Ground		450	75	100	11						
36	29	Nidoran??	Poison		275	55	47	5						
37	30	Nidorina	Poison		365	70	62	6						
38	31	Nidoqueer	Poison	Ground	505	90	92	8						
39	32	Nidoran??	Poison		273	46	57	4						
40	33	Nidorino	Poison		365	61	72	5						
41	34	Nidoking	Poison	Ground	505	81	102	7						

datasets_121_280_Pokemon

766	694	Helioptile	Electric	Normal	289	44	38	33	61	43	70	6	FALSE	
767	695	Heliolisk	Electric	Normal	481	62	55	52	109	94	109	6	FALSE	
768	696	Tyrunt	Rock	Dragon	362	58	89	77	45	45	48	6	FALSE	
769	697	Tyrantrum	Rock	Dragon	521	82	121	119	69	59	71	6	FALSE	
770	698	Amaura	Rock	Ice	362	77	59	50	67	63	46	6	FALSE	
771	699	Aurorus	Rock	Ice	521	123	77	72	99	92	58	6	FALSE	
772	700	Sylveon	Fairy		525	95	65	65	110	130	60	6	FALSE	
773	701	Hawlucha	Fighting	Flying	500	78	92	75	74	63	118	6	FALSE	
774	702	Dedenne	Electric	Fairy	431	67	58	57	81	67	101	6	FALSE	
775	703	Carbink	Rock	Fairy	500	50	50	150	50	150	50	6	FALSE	
776	704	Goomy	Dragon		300	45	50	35	55	75	40	6	FALSE	
777	705	Sliggoo	Dragon		452	68	75	53	83	113	60	6	FALSE	
778	706	Goodra	Dragon		600	90	100	70	110	150	80	6	FALSE	
779	707	Klefki	Steel	Fairy	470	57	80	91	80	87	75	6	FALSE	
780	708	Phantump	Ghost	Grass	309	43	70	48	50	60	38	6	FALSE	
781	709	Trevenant	Ghost	Grass	474	85	110	76	65	82	56	6	FALSE	
782	710	Pumpkabo	Ghost	Grass	335	49	66	70	44	55	51	6	FALSE	
783	710	Pumpkabo	Ghost	Grass	335	44	66	70	44	55	56	6	FALSE	
784	710	Pumpkabo	Ghost	Grass	335	54	66	70	44	55	46	6	FALSE	
785	710	Pumpkabo	Ghost	Grass	335	59	66	70	44	55	41	6	FALSE	
786	711	GourgeistA	Ghost	Grass	494	65	90	122	58	75	84	6	FALSE	
787	711	GourgeistS	Ghost	Grass	494	55	85	122	58	75	99	6	FALSE	
788	711	GourgeistL	Ghost	Grass	494	75	95	122	58	75	69	6	FALSE	
789	711	GourgeistS	Ghost	Grass	494	85	100	122	58	75	54	6	FALSE	
790	712	Bergmite	Ice		304	55	69	85	32	35	28	6	FALSE	
791	713	Avalugg	Ice		514	95	117	184	44	46	28	6	FALSE	
792	714	Noibat	Flying	Dragon	245	40	30	35	45	40	55	6	FALSE	
793	715	Noivern	Flying	Dragon	535	85	70	80	97	80	123	6	FALSE	
794	716	Xerneas	Fairy		680	126	131	95	131	98	99	6	TRUE	
795	717	Yveltal	Dark	Flying	680	126	131	95	131	98	99	6	TRUE	
796	718	Zygarde50	Dragon	Ground	600	108	100	121	81	95	95	6	TRUE	
797	719	Diancie	Rock	Fairy	600	50	100	150	100	150	50	6	TRUE	
798	719	DiancieMe	Rock	Fairy	700	50	160	110	160	110	110	6	TRUE	
799	720	HoopaHoc	Psychic	Ghost	600	80	110	60	150	130	70	6	TRUE	
800	720	HoopaHoc	Psychic	Dark	680	80	160	60	170	130	80	6	TRUE	
801	721	Volcanion	Fire	Water	600	80	110	120	130	90	70	6	TRUE	
802														
803														

You should be able to identify:

- what needs to be checked before running a mining algorithm.

Data understanding is part of modelling, not a separate admin step.

WHAT IS DATA? HOW TO UNDERSTANDING THE DATA?

COLLECTION OF *DATA OBJECTS* AND THEIR *ATTRIBUTES*

WHAT IS AN ATTRIBUTE? AN *ATTRIBUTE* IS A PROPERTY OR CHARACTERISTIC OF AN OBJECT

Examples: eye color of a person, temperature, etc.

Attribute is also known as variable, field, characteristic, dimension, or feature

WHAT IS OBJECT? A COLLECTION OF ATTRIBUTES DESCRIBE AN *OBJECT*

Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

ATTRIBUTE VALUES

ATTRIBUTE VALUES **ARE NUMBERS OR SYMBOLS ASSIGNED TO AN ATTRIBUTE FOR A PARTICULAR OBJECT**

DISTINCTION BETWEEN ATTRIBUTES AND ATTRIBUTE VALUES

Same attribute can be mapped to different attribute values

Example: height can be measured in feet or meters

Different attributes can be mapped to the same set of values

Example: Attribute values for ID and age are integers

But properties of attribute can be different than the properties of the values used to represent the attribute

TYPES OF ATTRIBUTES

THERE ARE DIFFERENT TYPES OF ATTRIBUTES

Nominal

Examples: ID numbers, eye color, zip codes

Ordinal

Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height {tall, medium, short}

Interval

Examples: calendar dates, temperatures in Celsius or Fahrenheit.

Ratio

Examples: temperature in Kelvin, length, counts, elapsed time (e.g., time to run a race)

PROPERTIES OF ATTRIBUTE VALUES

THE TYPE OF AN ATTRIBUTE DEPENDS ON WHICH OF THE FOLLOWING PROPERTIES/OPERATIONS IT POSSESSES:

Distinctness:	= ≠
Order:	< >
Differences are meaningful :	+ -
Ratios are meaningful	* /

Nominal attribute: distinctness

Ordinal attribute: distinctness & order

Interval attribute: distinctness, order & meaningful differences

Ratio attribute: all 4 properties/operations

	Attribute Type	Description	Examples	Operations
Categorical Qualitative	Nominal	Nominal attribute values only distinguish. (=, ≠)	zip codes, employee ID numbers, eye color, sex: {*male, female*}	mode, entropy, contingency correlation, $\chi 2$ test
	Ordinal	Ordinal attribute values also order objects. (<, >)	hardness of minerals, {*good, better, best*}, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Numeric Quantitative	Interval	For interval attributes, differences between values are meaningful. (+, -)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, *t* and *F* tests
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	temperature in Kelvin, monetary quantities, counts, age, mass, length, current	geometric mean, harmonic mean, percent variation

	Attribute Type	Transformation	Comments
Categorical Qualitative	Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
	Ordinal	An order preserving change of values, i.e., *new_value* = *f(old_value)* where *f* is a monotonic function	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by { 0.5, 1, 10}.
Numeric Quantitative	Interval	*new_value* = *a* * *old_value* + *b* where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
	Ratio	*new_value* = *a* * *old_value*	Length can be measured in meters or feet.

DISCRETE AND CONTINUOUS ATTRIBUTES

DISCRETE ATTRIBUTE

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables.
- Note: binary attributes are a special case of discrete attributes

CONTINUOUS ATTRIBUTE

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can only be measured and represented using a finite number of digits.

Continuous attributes are typically represented as floating-point variables.

TYPES OF DATA SETS

RECORD

Data Matrix

Document Data

Transaction Data

GRAPH

World Wide Web

Molecular Structures

ORDERED

Spatial Data

Temporal Data

Sequential Data

Genetic Sequence Data

RECORD DATA

DATA THAT CONSISTS OF A COLLECTION OF RECORDS, EACH OF WHICH CONSISTS OF A FIXED SET OF ATTRIBUTES

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

TRANSACTION DATA

A SPECIAL TYPE OF DATA, WHERE

Each transaction involves a set of items.

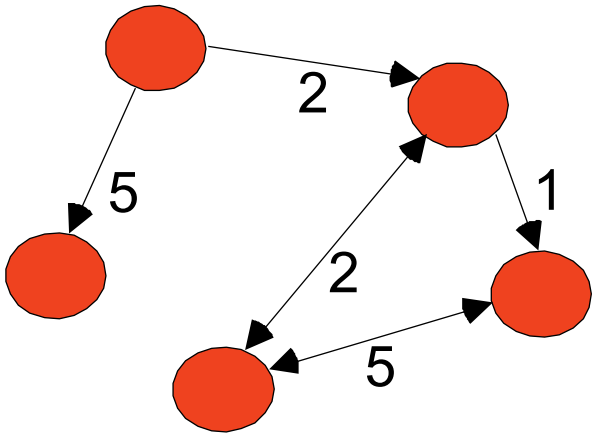
For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

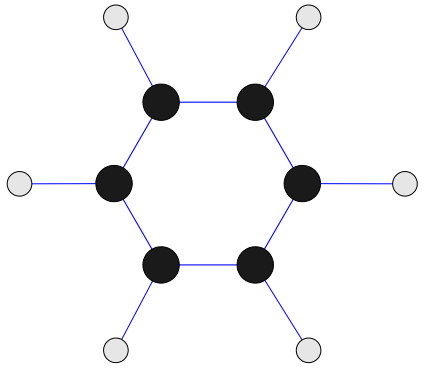
Can represent transaction data as record data

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

GRAPH DATA

EXAMPLES: GENERIC GRAPH, A MOLECULE, AND WEBPAGES





Benzene Molecule: C6H6

Useful Links:

- Bibliography
- Other Useful Web sites
  - ACM SIGKDD
  - KDnuggets
  - The Data Mine

Knowledge Discovery and Data Mining Bibliography

(Gets updated frequently, so visit often!)

- Books
- General Data Mining

Book References in Data Mining and Knowledge Discovery

Usama Fayyad, Gregory Piatetsky-Shapiro, Padhraic Smyth, and Ramasamy uthurasamy, "Advances in Knowledge Discovery and Data Mining", AAAI Press/the MIT Press, 1996.

J. Ross Quinlan, "C4.5: Programs for Machine Learning", Morgan Kaufmann Publishers, 1993.
Michael Berry and Gordon Linoff, "Data Mining Techniques (For Marketing, Sales, and Customer Support), John Wiley & Sons, 1997.

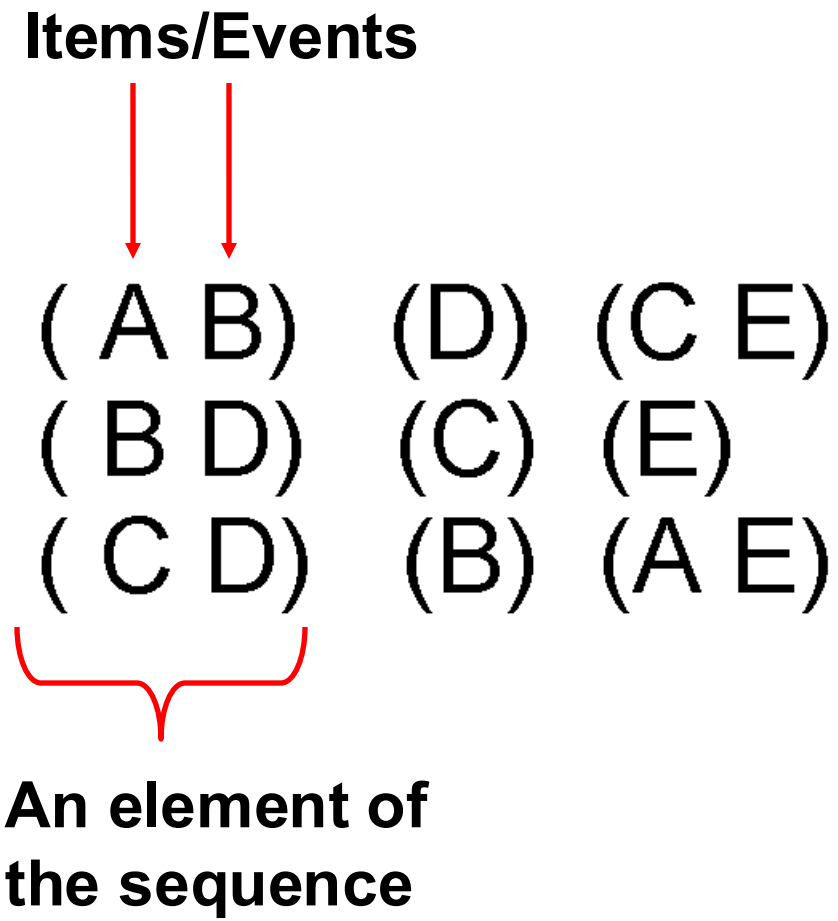
General Data Mining

Usama Fayyad, "Mining Databases: Towards Algorithms for Knowledge Discovery", Bulletin of the IEEE Computer Society Technical Committee on data Engineering, vol. 21, no. 1, March 1998.

Christopher Matheus, Philip Chan, and Gregory Piatetsky-Shapiro, "Systems for knowledge Discovery in databases", IEEE Transactions on Knowledge and Data Engineering, 5(6):903-913, December 1993.

ORDERED DATA

SEQUENCES OF TRANSACTIONS



ORDERED DATA

GENOMIC SEQUENCE DATA

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

DATA QUALITY

POOR DATA QUALITY NEGATIVELY AFFECTS MANY DATA PROCESSING EFFORTS

DATA MINING EXAMPLE: A CLASSIFICATION MODEL FOR DETECTING PEOPLE WHO ARE LOAN RISKS IS BUILT USING POOR DATA

Some credit-worthy candidates are denied loans

More loans are given to individuals that default

DATA QUALITY …

WHAT KINDS OF DATA QUALITY PROBLEMS?

HOW CAN WE DETECT PROBLEMS WITH THE DATA?

WHAT CAN WE DO ABOUT THESE PROBLEMS?

EXAMPLES OF DATA QUALITY PROBLEMS:

- Noise and outliers
- Wrong data
- Fake data
- Missing values
- Duplicate data

NOISE

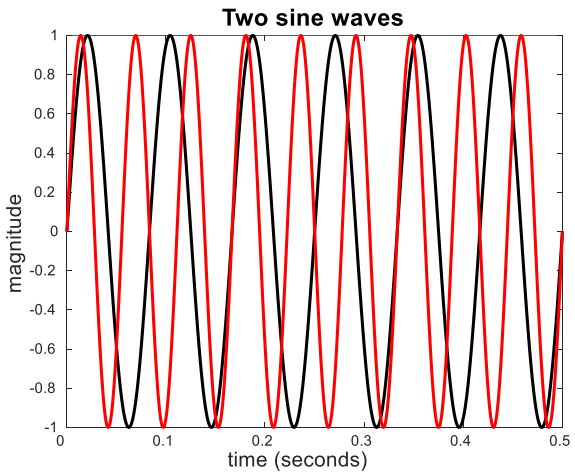
FOR OBJECTS, NOISE IS AN EXTRANEOUS OBJECT

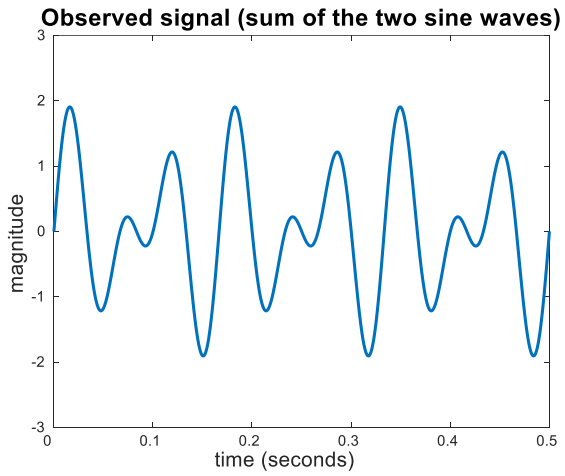
FOR ATTRIBUTES, NOISE REFERS TO MODIFICATION OF ORIGINAL VALUES

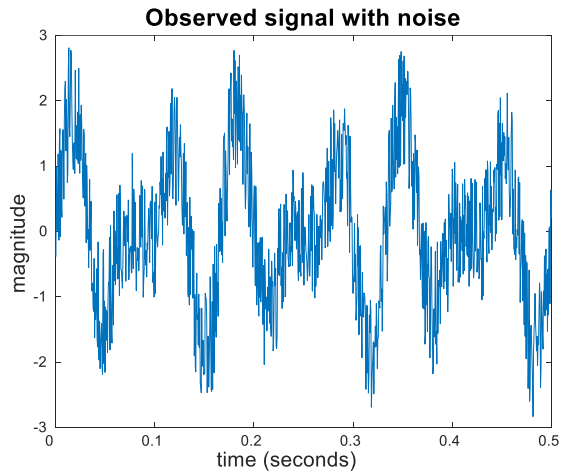
Examples: distortion of a person's voice when talking on a poor phone and "snow" on television screen

The figures below show two sine waves of the same magnitude and different frequencies, the waves combined, and the two sine waves with random noise

The magnitude and shape of the original signal is distorted







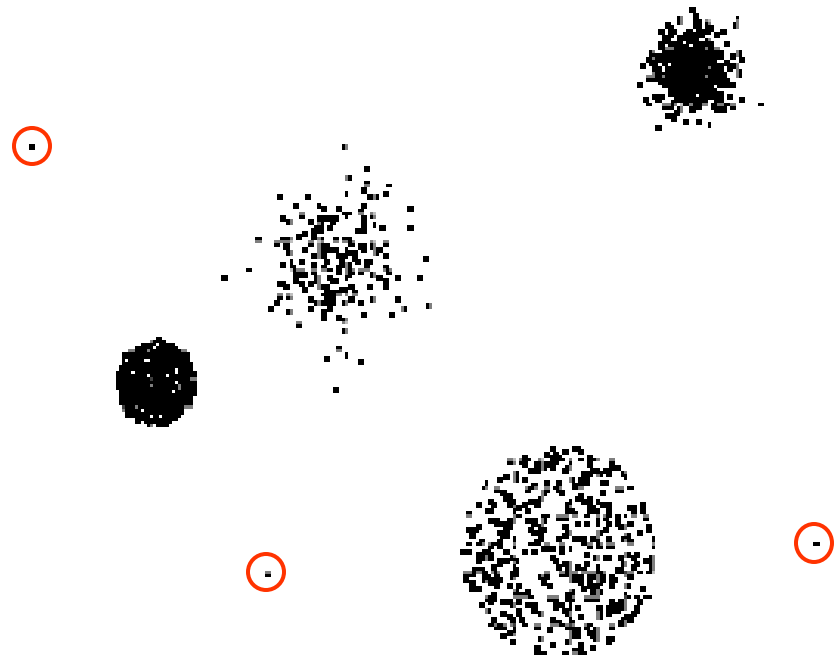
OUTLIERS

OUTLIERS **ARE DATA OBJECTS WITH CHARACTERISTICS THAT ARE CONSIDERABLY DIFFERENT THAN MOST OF THE OTHER DATA OBJECTS IN THE DATA SET**

Case 1: Outliers are noise that interferes with data analysis

Case 2: Outliers are the goal of our analysis

- Credit card fraud
- Intrusion detection



MISSING VALUES

REASONS FOR MISSING VALUES

Information is not collected
(e.g., people decline to give their age and weight)

Attributes may not be applicable to all cases
(e.g., annual income is not applicable to children)

HANDLING MISSING VALUES

Eliminate data objects or variables

Estimate missing values

- Example: time series of temperature
- Example: census results

Ignore the missing value during analysis

DUPLICATE DATA

DATA SET MAY INCLUDE DATA OBJECTS THAT ARE DUPLICATES, OR ALMOST DUPLICATES OF ONE ANOTHER

Major issue when merging data from heterogeneous sources

EXAMPLES:

Same person with multiple email addresses

DATA CLEANING

Process of dealing with duplicate data issues

DATA PREPROCESSING

AGGREGATION

SAMPLING

DISCRETIZATION AND BINARIZATION

ATTRIBUTE TRANSFORMATION

DIMENSIONALITY REDUCTION

FEATURE SUBSET SELECTION

FEATURE CREATION

AGGREGATION

COMBINING TWO OR MORE ATTRIBUTES (OR OBJECTS) INTO A SINGLE ATTRIBUTE (OR OBJECT)

PURPOSE

Data reduction - reduce the number of attributes or objects

Change of scale

- Cities aggregated into regions, states, countries, etc.
- Days aggregated into weeks, months, or years

More “stable” data - aggregated data tends to have less variability

EXAMPLE: PRECIPITATION IN AUSTRALIA

THIS EXAMPLE IS BASED ON PRECIPITATION IN AUSTRALIA FROM THE PERIOD 1982 TO 1993.

The next slide shows

A histogram for the standard deviation of average monthly precipitation for 3,030 0.5◦ by 0.5◦ grid cells in Australia, and

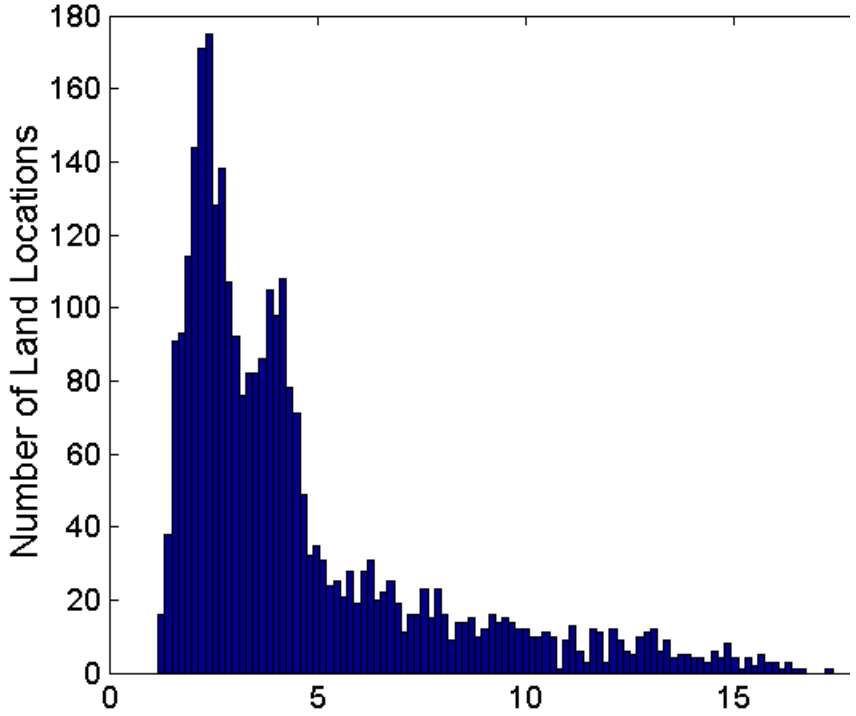
A histogram for the standard deviation of the average yearly precipitation for the same locations.

THE AVERAGE YEARLY PRECIPITATION HAS LESS VARIABILITY THAN THE AVERAGE MONTHLY PRECIPITATION.

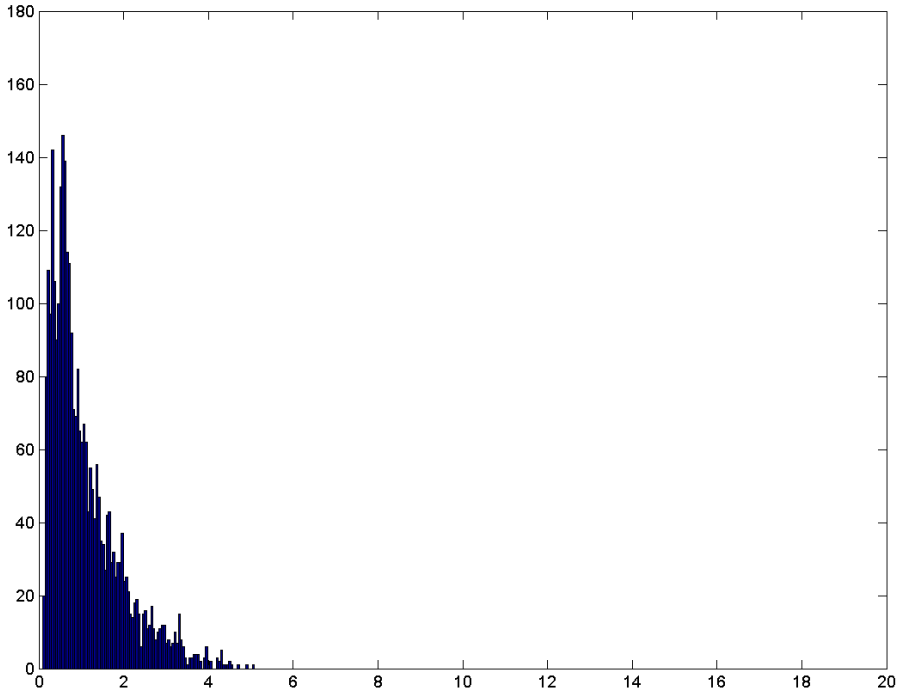
ALL PRECIPITATION MEASUREMENTS (AND THEIR STANDARD DEVIATIONS) ARE IN CENTIMETERS.

EXAMPLE: PRECIPITATION IN AUSTRALIA …

Variation of Precipitation in Australia



Standard Deviation of Average Monthly Precipitation



Standard Deviation of Average Yearly Precipitation

SAMPLING

SAMPLING IS THE MAIN TECHNIQUE EMPLOYED FOR DATA REDUCTION.

It is often used for both the preliminary investigation of the data and the final data analysis.

STATISTICIANS OFTEN SAMPLE BECAUSE OBTAINING THE ENTIRE SET OF DATA OF INTEREST IS TOO EXPENSIVE OR TIME CONSUMING.

SAMPLING IS TYPICALLY USED IN DATA MINING BECAUSE PROCESSING THE ENTIRE SET OF DATA OF INTEREST IS TOO EXPENSIVE OR TIME CONSUMING.

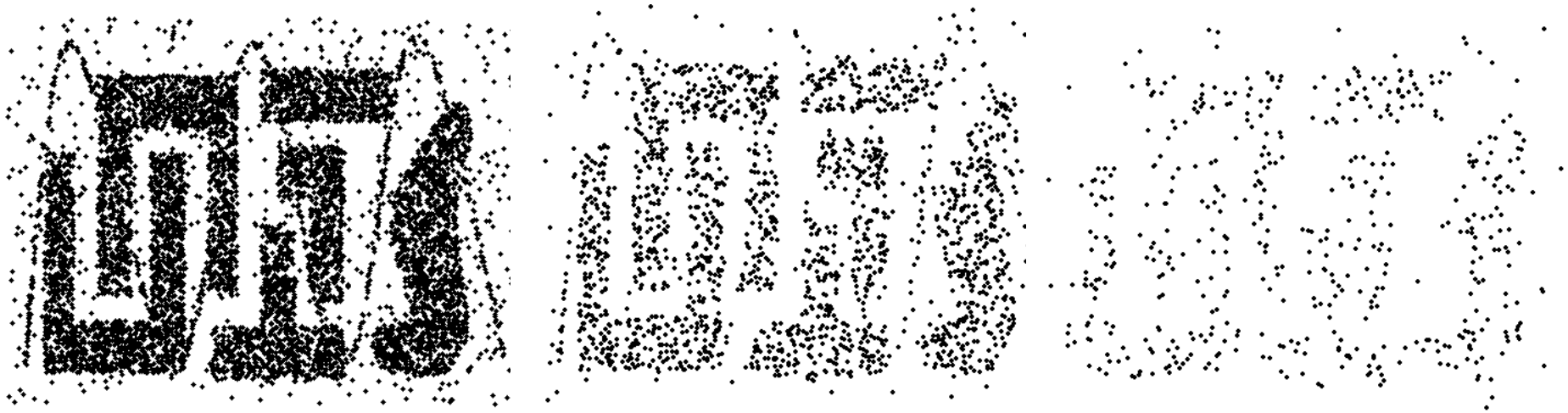
SAMPLING …

THE KEY PRINCIPLE FOR EFFECTIVE SAMPLING IS THE FOLLOWING:

Using a sample will work almost as well as using the entire data set, if the sample is representative

A sample is representative if it has approximately the same properties (of interest) as the original set of data

SAMPLE SIZE



8000 points

2000 Points

500 Points

TYPES OF SAMPLING

SIMPLE RANDOM SAMPLING

There is an equal probability of selecting any particular item

Sampling without replacement

As each item is selected, it is removed from the population

Sampling with replacement

Objects are not removed from the population as they are selected for the sample.

In sampling with replacement, the same object can be picked up more than once

When do we need Sampling with replacement?
e.g. Statistical needs: Each data point is required to be drawn independently of others to satisfy the assumption that

- Each data point is drawn independently of others,
- All data points come from the same distribution.

STRATIFIED SAMPLING

Split the data into several partitions; then draw random samples from each partition

CURSE OF DIMENSIONALITY

WHEN DIMENSIONALITY INCREASES, DATA BECOMES INCREASINGLY SPARSE IN THE SPACE THAT IT OCCUPIES

DEFINITIONS OF DENSITY AND DISTANCE BETWEEN POINTS, WHICH ARE CRITICAL FOR CLUSTERING AND OUTLIER DETECTION, BECOME LESS MEANINGFUL

DIMENSIONALITY REDUCTION

PURPOSE:

Avoid curse of dimensionality

Reduce amount of time and memory required by data mining algorithms

Allow data to be more easily visualized

May help to eliminate irrelevant features or reduce noise

TECHNIQUES

Principal Components Analysis (PCA)

Singular Value Decomposition

Others: supervised and non-linear techniques

FEATURE CREATION

CREATE NEW ATTRIBUTES THAT CAN CAPTURE THE IMPORTANT INFORMATION IN A DATA SET MUCH MORE EFFICIENTLY THAN THE ORIGINAL ATTRIBUTES

THREE GENERAL METHODOLOGIES:

Feature extraction

Example: extracting edges from images

Feature construction

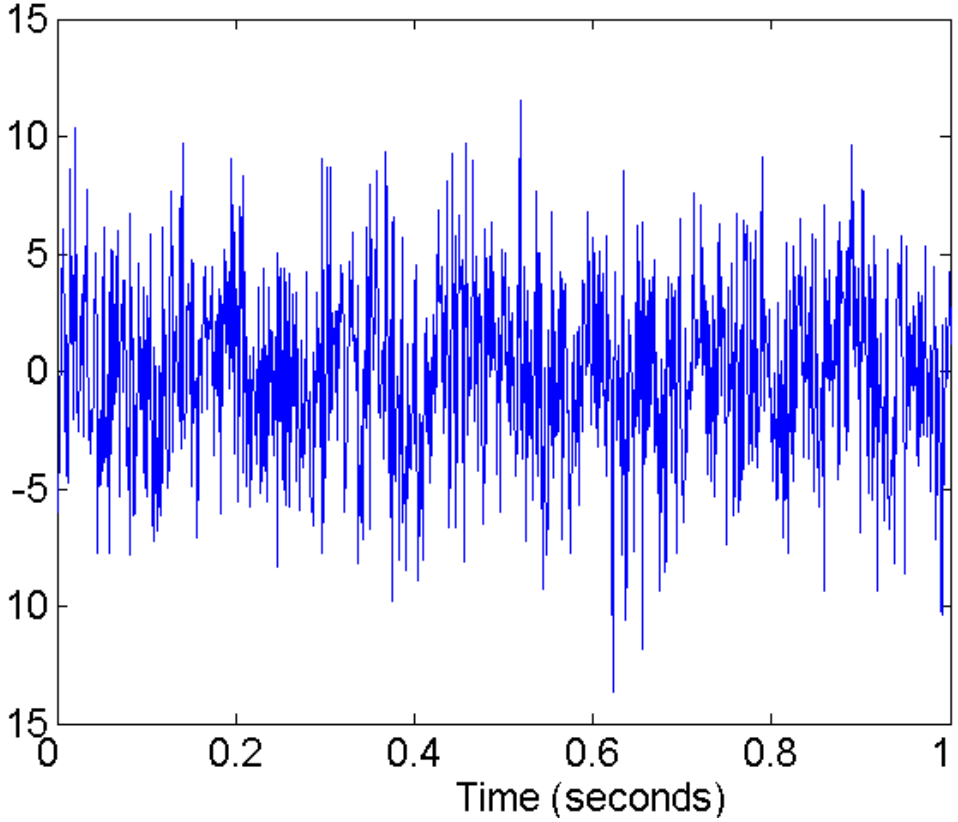
Example: dividing mass by volume to get density

Mapping data to new space

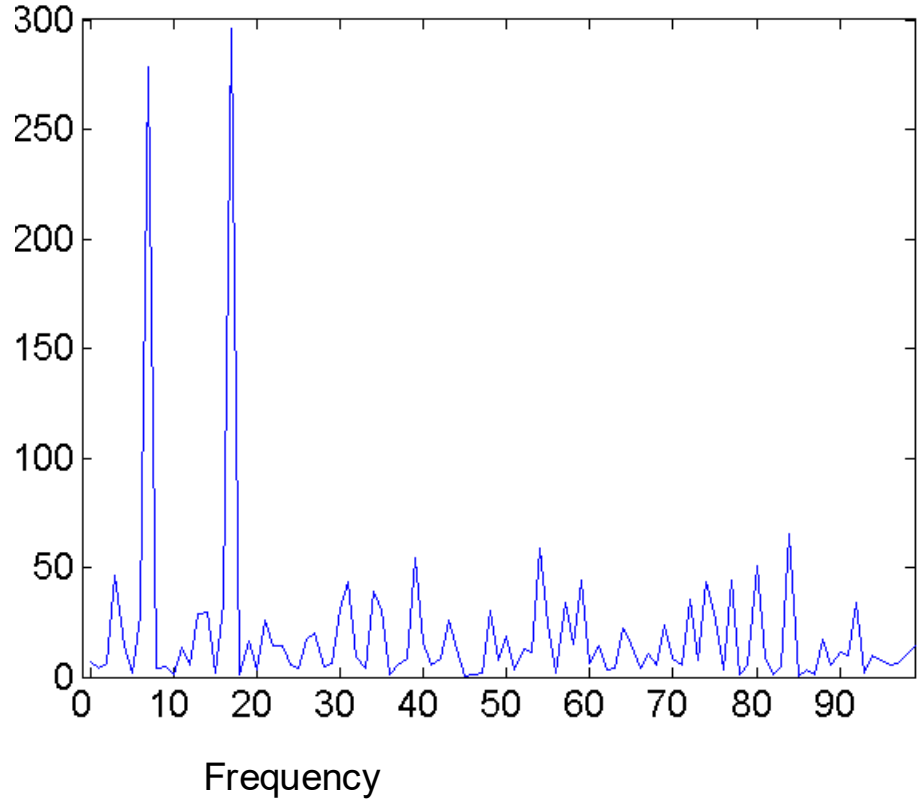
Example: Fourier and wavelet analysis

MAPPING DATA TO A NEW SPACE

- Fourier and wavelet transform



Two Sine Waves + Noise



Frequency

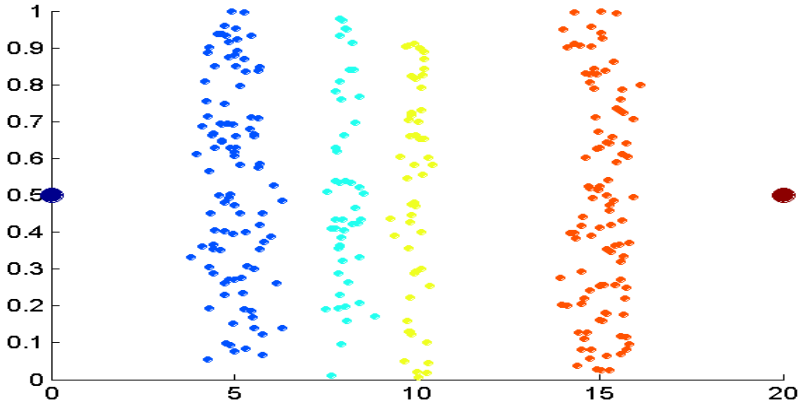
DISCRETIZATION

DISCRETIZATION IS THE PROCESS OF CONVERTING A CONTINUOUS ATTRIBUTE INTO AN ORDINAL ATTRIBUTE

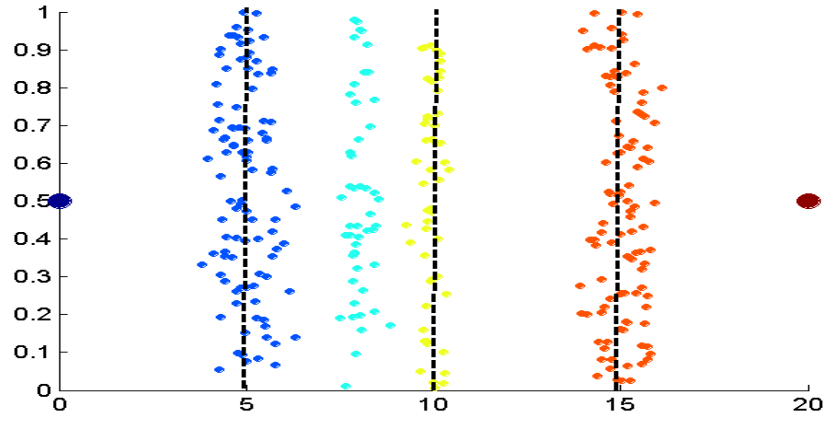
A potentially infinite number of values are mapped into a small number of categories

Discretization is used in both unsupervised and supervised settings

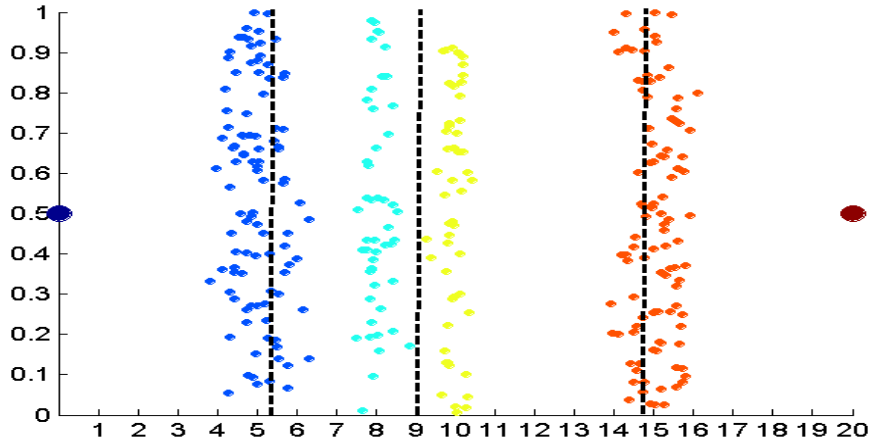
UNSUPERVISED DISCRETIZATION (WITHOUT USING CLASS LABELS)



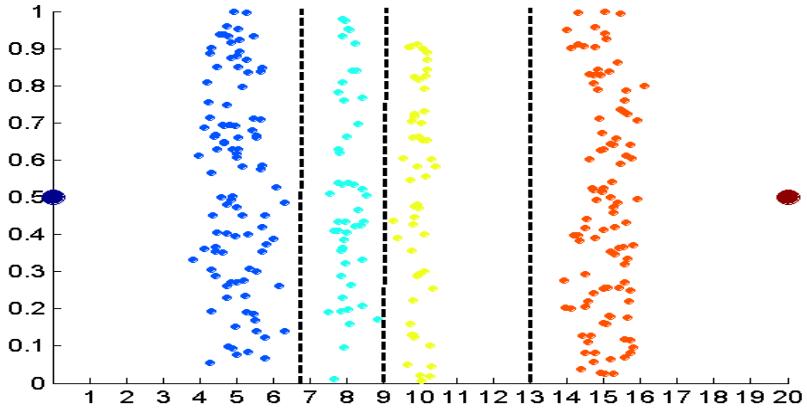
Data consists of four groups of points and two outliers.



Equal interval width approach used to obtain 4 values.



Equal frequency approach used to obtain 4 values.

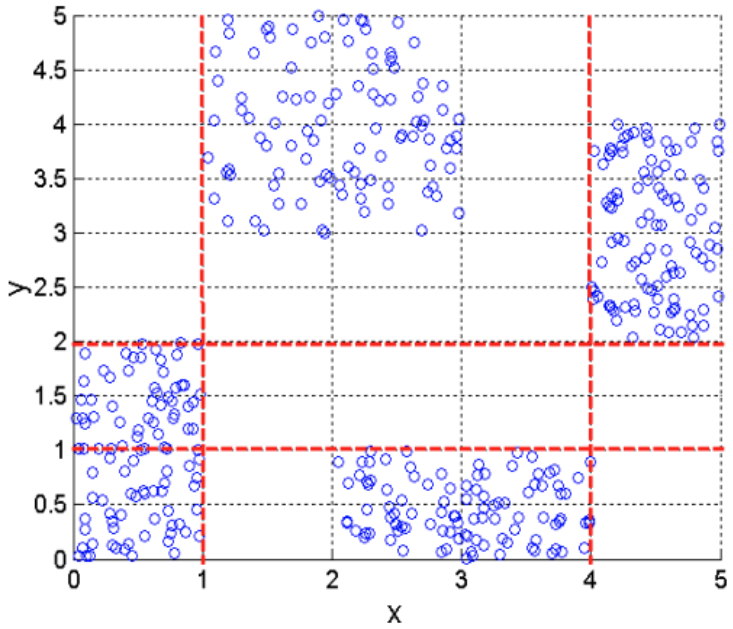


K-means approach to obtain 4 values.

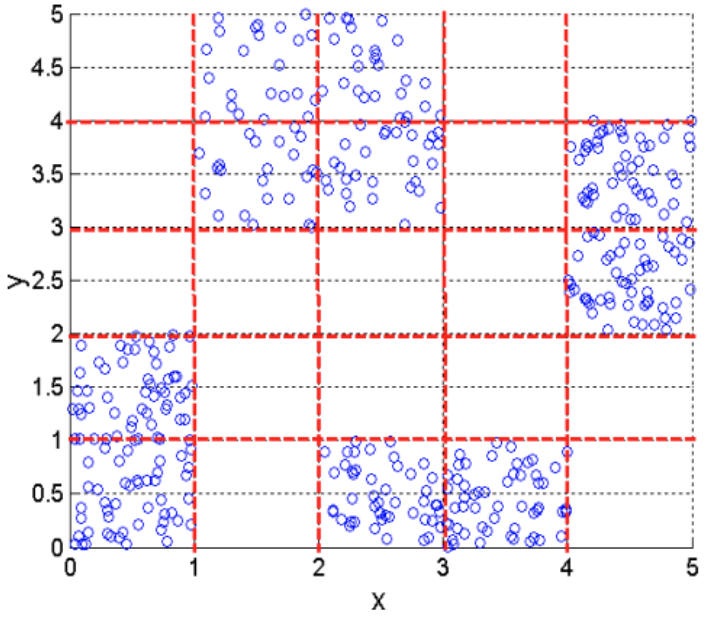
DISCRETIZATION IN SUPERVISED SETTINGS (USING CLASS LABELS)

Many classification algorithms work best if both the independent and dependent variables have only a few values

We give an illustration of the usefulness of discretization using the following example.



3 categories for both x and y



5 categories for both x and y

BINARIZATION

BINARIZATION MAPS A CONTINUOUS OR CATEGORICAL ATTRIBUTE INTO ONE OR MORE BINARY VARIABLES

- Also known as one-hot encoding

Table 2.6. Conversion of a categorical attribute to five asymmetric binary attributes.

Categorical Value	Integer Value	x_1	x_2	x_3	x_4	x_5
awful	0	1	0	0	0	0
poor	1	0	1	0	0	0
OK	2	0	0	1	0	0
good	3	0	0	0	1	0
great	4	0	0	0	0	1

ATTRIBUTE TRANSFORMATION

AN ATTRIBUTE TRANSFORM IS A FUNCTION THAT MAPS THE ENTIRE SET OF VALUES OF A GIVEN ATTRIBUTE TO A NEW SET OF REPLACEMENT VALUES SUCH THAT EACH OLD VALUE CAN BE IDENTIFIED WITH ONE OF THE NEW VALUES

Simple functions: x^k, $\log(x)$, e^x, $|x|$

Normalization

- Refers to various techniques to adjust to differences among attributes in terms of frequency of occurrence, mean, variance, range
- Take out unwanted, common signal, e.g., seasonality

In statistics, standardization refers to subtracting off the means and dividing by the standard deviation

SIMILARITY AND DISSIMILARITY MEASURES

SIMILARITY MEASURE

Numerical measure of how alike two data objects are.

Is higher when objects are more alike.

Often falls in the range [0,1]

DISSIMILARITY MEASURE

Numerical measure of how different two data objects are

Lower when objects are more alike

Minimum dissimilarity is often 0

Upper limit varies

PROXIMITY REFERS TO A SIMILARITY OR DISSIMILARITY

SIMILARITY/DISSIMILARITY FOR SIMPLE ATTRIBUTES

The following table shows the similarity and dissimilarity between two objects, x and y, with respect to a single, simple attribute.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = \lvert x - y \rvert / (n-1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = \lvert x - y \rvert$	$s = -d$, $s = \frac{1}{1+d}$, $s = e^{-d}$, $s = 1 - \frac{d - min_d}{max_d - min_d}$

EUCLIDEAN DISTANCE

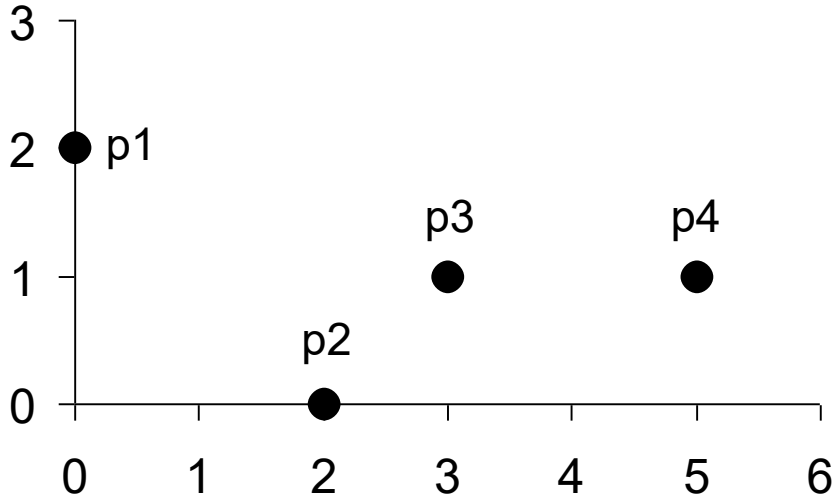
EUCLIDEAN DISTANCE

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

where *n* is the number of dimensions (attributes) and x_k and y_k are, respectively, the kth attributes (components) or data objects $\mathbf{x}$ and $\mathbf{y}$.

- Standardization is necessary, if scales differ.

EUCLIDEAN DISTANCE



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

FEATURE SUBSET SELECTION

ANOTHER WAY TO REDUCE DIMENSIONALITY OF DATA

REDUNDANT FEATURES

Duplicate much or all of the information contained in one or more other attributes

Example: purchase price of a product and the amount of sales tax paid

IRRELEVANT FEATURES

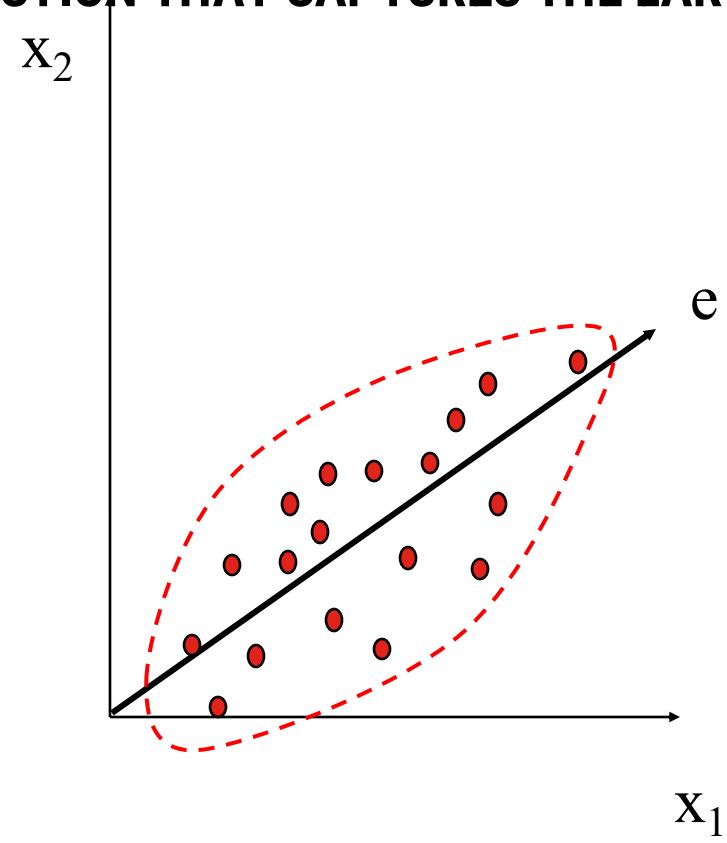
Contain no information that is useful for the data mining task at hand

Example: students' ID is often irrelevant to the task of predicting students' GPA

MANY TECHNIQUES DEVELOPED, ESPECIALLY FOR CLASSIFICATION

DIMENSIONALITY REDUCTION: PCA

GOAL IS TO FIND A PROJECTION THAT CAPTURES THE LARGEST AMOUNT OF VARIATION IN DATA



SUMMARY

ATTRIBUTES AND OBJECTS

- Nominal, Ordinal. Interval. Ratio
- Discrete, continuous

TYPES OF DATA

- Record
- Graph
- ordered

DATA QUALITY

- Noise and outliers
- Wrong data
- Missing values
- Duplicate data

SIMILARITY AND DISTANCE

DATA PREPROCESSING

- Aggregation
- Sampling
- Discretization and Binarization
- Attribute Transformation
- Dimensionality Reduction
- Feature subset selection
- Feature creation

LABS

- Wed 11:00-13:00, BG103 (Jianqiao: jianqiao.zhang@latrobe.edu.au)
- Wed 13:00-15:00, BG103 (Jianqiao: jianqiao.zhang@latrobe.edu.au)
- Thu 11:00-13:00, BG 103 (Tom: hoangnguyen.nguyen@latrobe.edu.au)

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THANK YOU

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